



Retinal Review Workbench

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Document control

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Version	Date	Change
1.0	2026-09-23	Rebuilt as a concise Quarto manual for the current clinician workflow.

1.1 Quick start

Use this sequence for an ordinary supported retinal image:

1. In **Worklist**, select the case and choose **Confirm Image**.
2. In **Review**, inspect the retinal image. Choose **Analyze** only when optional model assistance is useful.
3. Use **Show AI suggestions** and the lesion filter to inspect PRISM overlays. No individual ROI confirmation is required.
4. In **Clinician Review**, select the final grade and choose **Confirm DR Grade**.
5. Open **Edit Annotations** only when a correction or human annotation is needed. Choose **Confirm Annotation** after reviewing the active set.
6. In **Datasets**, check **DR-ready** and **Lesion-ready** separately before export.

IMPORTANT - three confirmation milestones

Confirm Image, **Confirm DR Grade**, and **Confirm Annotation** record separate case-level milestones. AI suggestions are optional assistance. They do not have to be confirmed or rejected one by one before the clinical review can be completed.

CLINICIAN WORKFLOW

From admitted image to reviewed dataset

Model output is optional evidence. Human review remains authoritative.



The optional annotation step does not create a mandatory per-lesion confirmation queue.

Figure 1.1: Clinician workflow

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Before you begin

2.1 Intended users

This manual is for clinicians and reviewers using the local review workstation. Dataset export and hospital Model API administration are described only at the level needed to understand the clinician workflow; operator commands are documented separately in the repository deployment guides.

2.2 Scope

Retinal Review Workbench is a clinician-controlled research and clinical-support workstation. It keeps source identity, patient and eye context, human decisions, annotations, model evidence, and export provenance together for review.

It does not autonomously establish a diagnosis, disease truth, referral decision, or treatment decision. The final clinician review remains authoritative.

2.3 Supported input intent

The workstation accepts JPEG, PNG, TIFF, and supported single-frame ophthalmic fundus DICOM. Supported ophthalmic DICOM follows the existing display and provenance path.

MRI, OCT, PACS objects, non-fundus DICOM, unsupported SOP classes, multi-frame objects, and untested transfer syntaxes are outside model support. A recognized non-fundus DICOM is reported as **Unsupported modality**, not as a generic image-read failure.

2.4 Model-assisted evidence

RETFound provides a five-class DR grade suggestion through `retfound-aptos5`. PRISM-DR provides optional lesion suggestions through `prism-dr-5fold`. These outputs assist inspection; they do not replace clinician review.

A displayed confidence or score is a model score, not a calibrated clinical probability. An empty PRISM result does not prove that lesions are absent.

2.5 Workstation areas

Area	Use
Worklist	Scan cases, review admission status, resolve patient and eye context, and enter a case.
Review	Inspect the image and optional model assistance.
Clinician Review	Record the final DR grade or escalate an exception.
Edit Annotations	Add human annotations or make optional AI ROI class/removal changes.
Datasets	Inspect task-specific readiness and export the manifest package.
Models & Audit	Read-only model, inference, hash, and annotation provenance.
Settings	Select a Workspace, set the default reviewer, and manage next-action hints.

WARNING - human authority

Treat every model output as assistive evidence. Do not use a model score or an empty lesion result as a diagnosis, a referral decision, or proof that disease is absent.

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Worklist and case setup

3.1 Start a review session

Purpose. Open the workstation and select the Workspace that contains the cases for the current review session.

Procedure.

1. Start the review workstation using the local launcher provided by the operator.
2. Open <http://127.0.0.1:8000/app/>.
3. Open **Settings** and select or create the approved Workspace.
4. Return to **Worklist** and choose **Scan input folder**.

Expected result. The Worklist shows admitted files with plain-language status and available actions. Original source files remain unchanged.

3.2 Confirm Image

Purpose. Confirm the image context before analysis or downstream clinical review.

Before you begin. Open a case with **Confirm Image** available. Filename or resolver suggestions are evidence only until you confirm them.

Procedure.

1. Choose **Confirm Image** in the case action menu.
2. Review the displayed filename and admission status.
3. Enter or clear **Pseudonymous patient key**. Leave it empty when the case should remain unlinked.
4. Set **Eye** to **Left**, **Right**, or **Unknown**.
5. Enter **Reviewer name**, or use the workstation default when appropriate.
6. Choose **Confirm Image**.

Expected result. The patient and eye context, reviewer, timestamp, and image-confirmation milestone are saved. A supported fundus image can proceed to **Review**.

IMPORTANT - reviewer preference

The default reviewer is a browser-local convenience, not an authenticated identity. On a shared workstation, change or clear it when the user changes. Previously saved reviewer names are not

rewritten.

Worklist

Public & synthetic cases for clinician review

WARNINGS

Admitted retinal images for clinician review

Worklist

Review admitted images, resolve patient and eye identity, and keep AI assistance separate from the clinician decision.

Scan input folder

Refresh

Search filename or patient key

Filters

Sort: Filename

Cases

Patients

Search by filename or pseudonymous patient key.

11 cases - 11 need attention

IMAGE	PATIENT / EYE	AI RESULT	REVIEW	ACTION
 01_dr.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...
 02_dr.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...
 03_dr.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...
 04_dr.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...
 05_dr.jpg	Patient not linked Eye side needs confirmation Link patient and confirm eye	Not analyzed	Pending review	Review ...
	Patient not linked			

Figure 3.1: Worklist overview

3.3 Admission exceptions

Use the visible case status to decide the next action. A recognized MRI or other non-fundus DICOM remains **Unsupported modality**. Do not send it to RETFound or PRISM-DR.

CAUTION - source files

Do not rename, move, overwrite, or recompress admitted source images from the review workstation. Work with the recorded source and derived display paths.

4

Review

4.1 Inspect the image

Purpose. Inspect the original retinal image and decide whether optional model evidence is useful.

Before you begin. Confirm the image from **Worklist**. The case must be a supported retinal input.

Procedure.

1. Open the case in **Review**.
2. Use the viewer controls to fit, zoom, pan, inspect coordinates, or open a larger view.
3. Read the patient and eye context above the image.
4. Choose **Analyze** when model assistance is needed and the Model API is ready.
5. Use **Show AI suggestions** to show or hide PRISM overlays.
6. Use **Filter lesion overlays** to choose **All lesion classes**, **MA - Microaneurysm**, **HE - Hemorrhage**, **EX - Hard exudate**, or **SE - Soft exudate**.

Expected result. The retinal image remains the primary review surface. The right side presents a compact DR assessment, lesion suggestion summary, and model/runtime status.

4.2 Read model suggestions

RETFound presents a suggested DR grade and model score. PRISM-DR presents raw lesion suggestions by class. Untouched overlays may show a class and score such as HE - 0.90.

The lesion summary describes AI suggestions. It is not a queue of unresolved clinician tasks. Review completion does not depend on confirming, rejecting, or reducing an AI detection count to zero.

4.3 Models & Audit

Open **Models & Audit** when technical inspection is needed. This read-only page can show model IDs and revisions, runtime state, inference metadata, source and analysis hashes, detection counts, and annotation provenance. It is an audit surface, not another review queue.

NOTE - score meaning

If a clinician changes an AI lesion class in **Edit Annotations**, the original AI score remains attached to the original class. The corrected human class must not be displayed with that score.

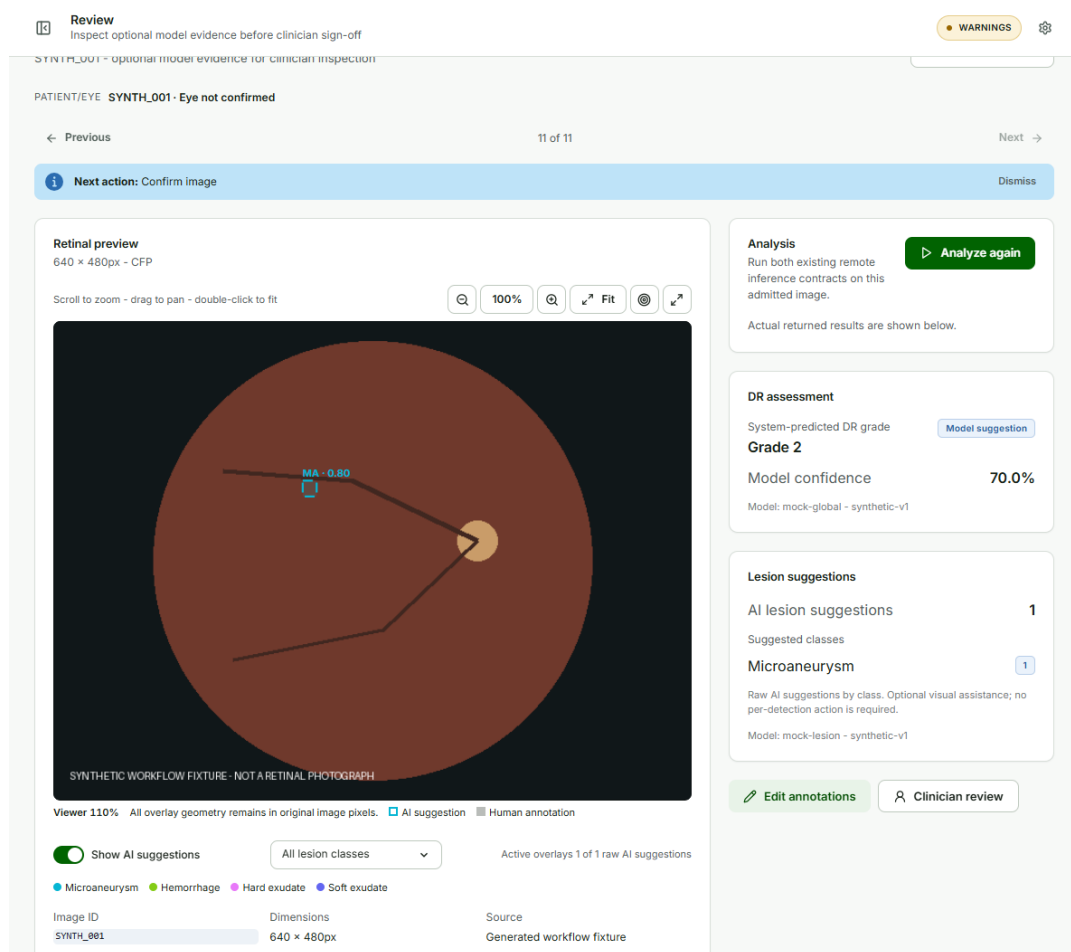


Figure 4.1: Review image and assistance

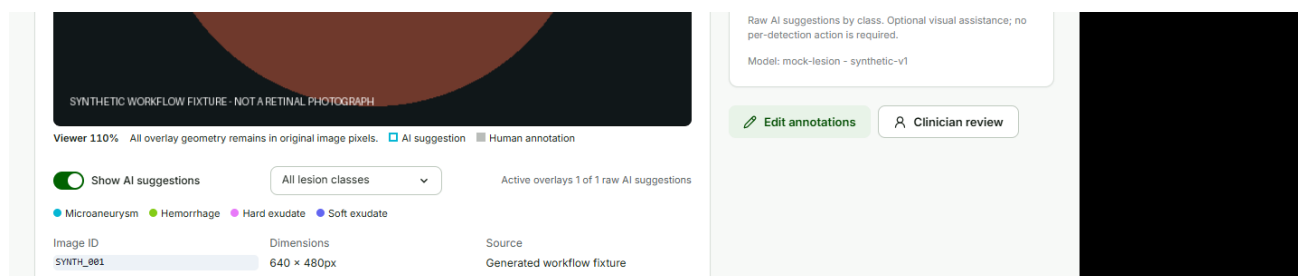


Figure 4.2: Overlay controls

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Clinician decisions and annotations

5.1 Confirm DR Grade

Purpose. Record the clinician's final DR grade.

Procedure.

1. Open **Clinician Review**.
2. Review the optional **AI suggestion**.
3. Select a value from **Final DR grade**.
4. Check or edit **Reviewer name**.
5. Add a **Remark** when useful.
6. Choose **Confirm DR Grade**. Choose **Escalate** for an exception path.

Expected result. The human grade is saved as the authoritative clinician decision. A matching AI suggestion is recorded as `AI_ACCEPTED`, a different grade as `AI_CORRECTED`, and a grade without an AI result as `MANUAL`.

5.2 Edit Annotations

Purpose. Add or correct the active annotation set when the clinician identifies a change that needs to be recorded.

Procedure.

1. Choose **Edit annotations** from the case review surface.
2. Keep **AI suggestions** shown when comparing model evidence, or hide them to reduce visual density.
3. For a human annotation, select a tool, choose the canonical lesion class, and draw on the image.
4. Use the available selection and geometry controls for human annotations when needed.
5. Choose **Save draft** to persist work without confirming the set.
6. Choose **Confirm Annotation** after reviewing the active set.

Expected result. Human annotations remain separate from original AI evidence. A later edit invalidates the previous annotation confirmation until the current active set is confirmed again.

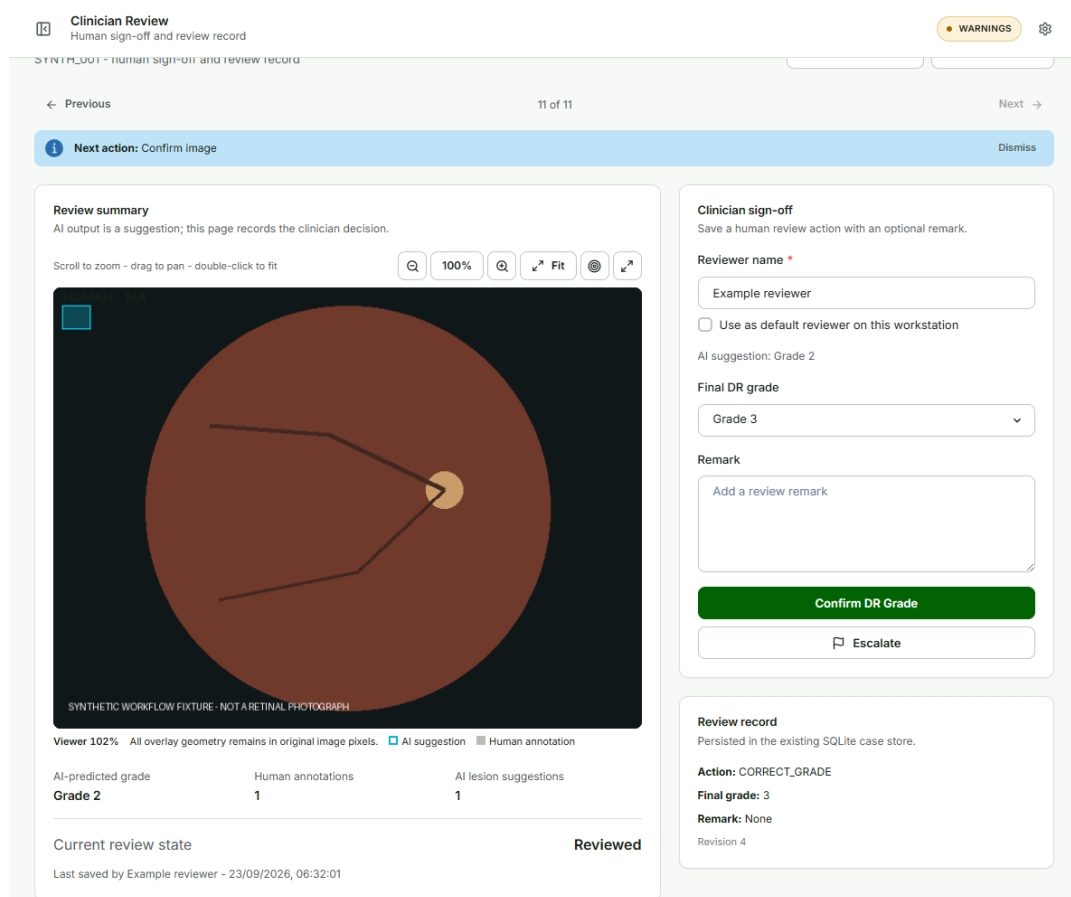


Figure 5.1: Grade confirmation

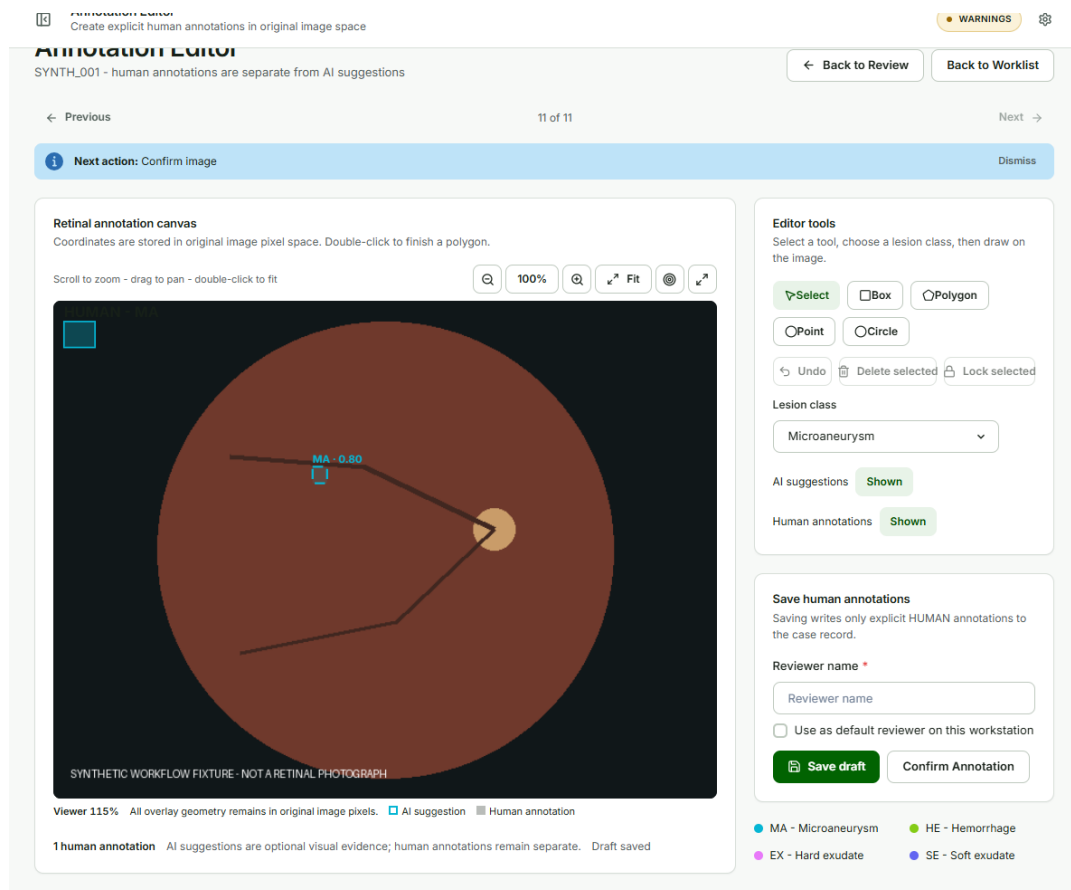


Figure 5.2: Annotation editor

5.3 Correct or remove one AI ROI

AI ROI editing is optional. Use it only when an obvious model class or detection needs correction.

1. In **Edit Annotations**, show **AI suggestions**.
2. Select the AI ROI on the image.
3. Read the original **AI suggestion**, including its original class and score.
4. Choose a different value under **Change lesion class**, then choose **Save class correction**, or choose **Remove this detection**.
5. Enter the actual **Reviewer name** used for the change.

The corrected overlay uses the reviewed class and indicates a clinician correction without assigning the AI score to that class. A removed detection is excluded from the active reviewed overlay while its original identity, geometry, score, model, and provenance remain available for audit.

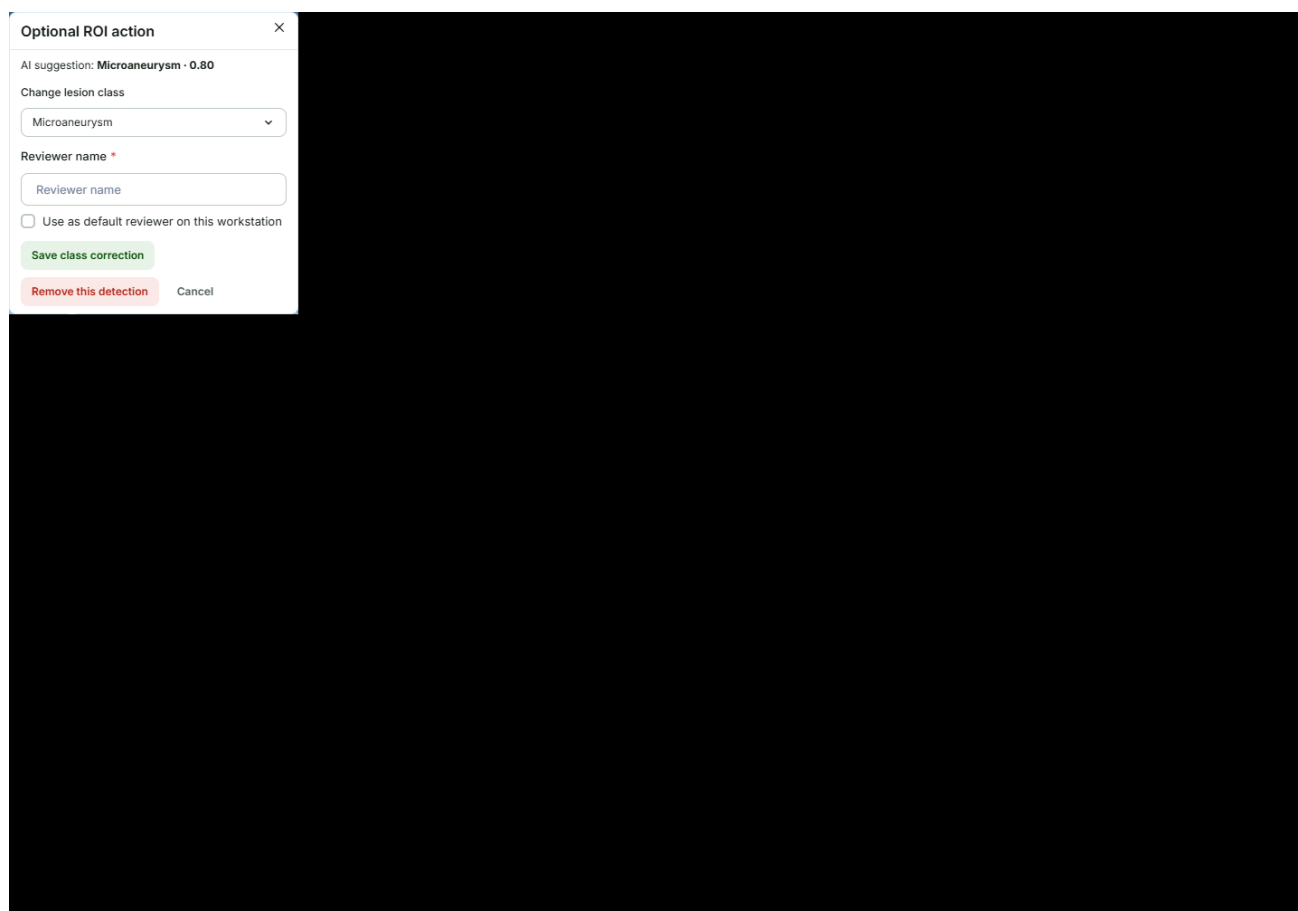


Figure 5.3: ROI action

WARNING - preserve score meaning

If the AI suggestion is Hemorrhage - 0.90 and the clinician changes the reviewed class to Microaneurysm, the result is **Microaneurysm - Clinician corrected**. It must not be shown as Microaneurysm - 0.90.

Geometry-level correction remains available through the existing optional CVAT workflow. This page does not replace CVAT with a second dense geometry editor.

5.4 Default reviewer

Purpose. Reduce repeated typing on a shared review workstation.

1. Open **Settings**.
2. Enter a name in **Default reviewer**.
3. Clear or replace the value when the workstation user changes.

New or unattributed review forms are prefilled. The reviewer remains editable and can be cleared. Existing recorded reviewers are never rewritten by a later preference change. The preference is not stored in patient or image metadata and does not prove identity.

5.5 Save and move between cases

Use **Previous** and **Next** to move between cases. Navigation does not save or confirm changes. In **Edit Annotations**, use **Save & Next Case** when the current draft should be saved before opening the next case.

If changes are unsaved, saving, or have failed, the application may show a navigation guard. Follow the prompt and retry **Save draft** when needed. **Next action** is guidance only; it does not change clinical state.

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Datasets and audit

6.1 Check readiness

Purpose. Check task-specific readiness without mixing DR-grade and lesion-annotation rules.

1. Open **Datasets**.
2. Choose **All**, **DR-ready**, **Lesion-ready**, **Needs review**, or **Excluded**.
3. Read the separate DR and lesion status in each case row.
4. Correct the case in **Clinician Review** or **Edit Annotations** when the status requires it.

Expected result. Classification and lesion readiness remain separate. A final grade can make a case DR-ready without annotation confirmation. A confirmed annotation set can make a case Lesion-ready without a final DR grade, subject to the existing dataset rules.

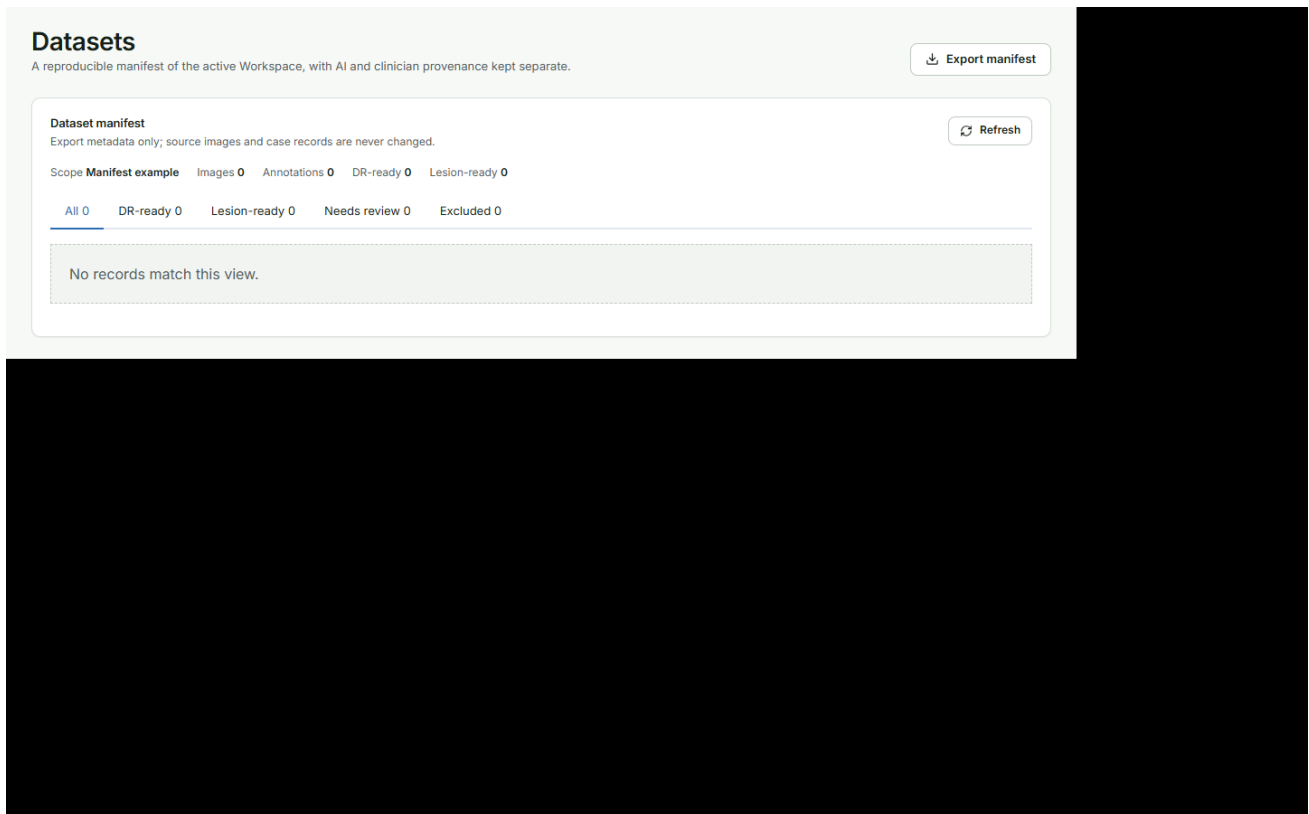


Figure 6.1: Dataset readiness

6.2 Export the manifest

1. Open an active Workspace.
2. Verify the current case state and readiness.
3. Choose **Export manifest**.
4. Confirm the success message and output location shown by the application.

The export package contains `manifest.json`, `images.csv`, and `annotations.csv`. The package uses schema `s4.dataset-manifest.v2` and keeps source identity, coordinate space, confirmation state, task readiness, and annotation provenance.

AI-only evidence is not silently promoted to ground truth. Untouched AI, case-level confirmation, human correction, human-authored, removed, and CVAT-imported states remain distinguishable through the export fields.

6.3 Read-only audit

Use **Models & Audit** to inspect:

- RETFound and PRISM-DR model IDs, revisions, runtime state, and limitations;
- inference metadata and lesion counts by class;
- source SHA-256 and analysis or derivative SHA-256 values;
- original AI evidence and annotation correction/removal provenance.

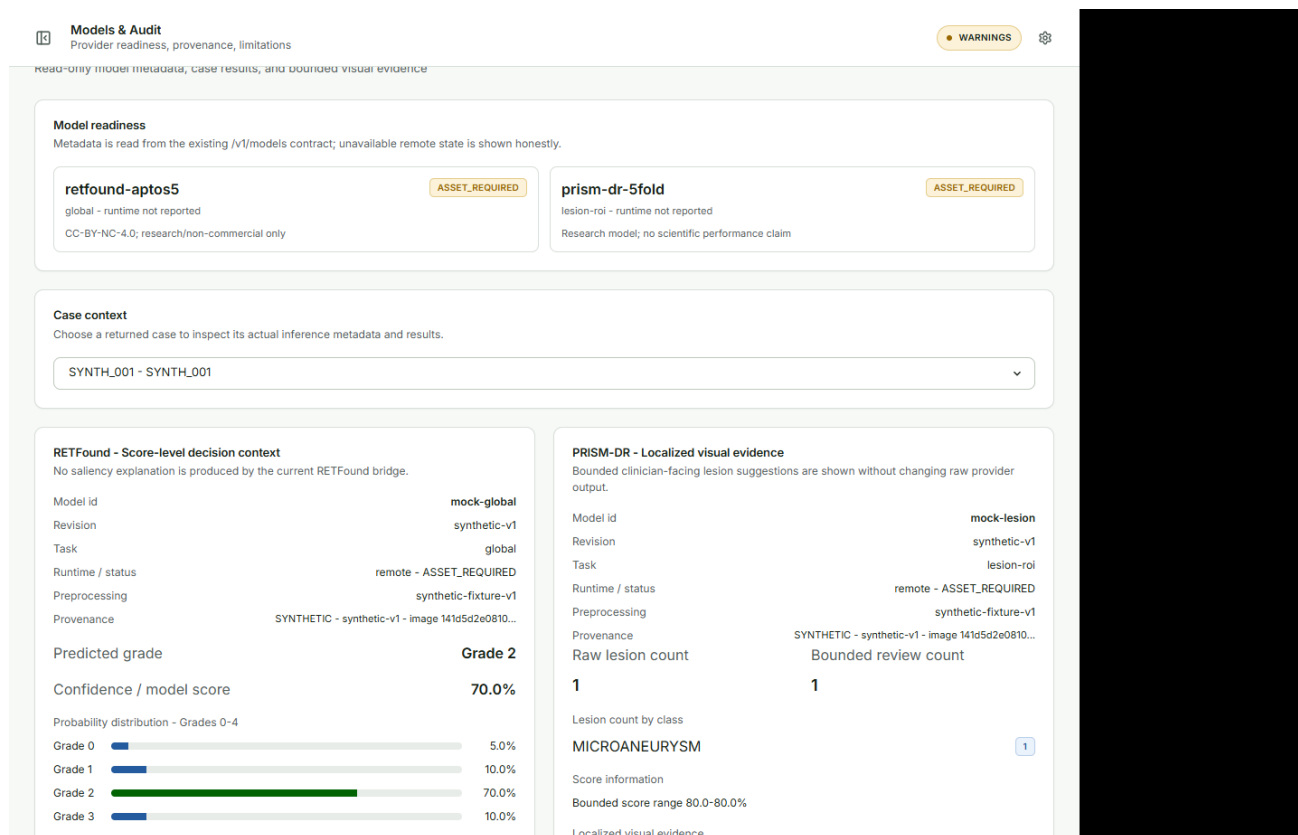


Figure 6.2: Models and audit

6.4 Optional CVAT

Use the configured CVAT Online integration for dense annotation or geometry correction. The existing round trip preserves source and prediction hashes and remains optional. CVAT Online is not part of the offline workstation requirement.

7

Troubleshooting and safety

7.1 Common situations

Observation	Action
Unsupported modality	The DICOM is recognized but is not supported ophthalmic fundus input. Do not send it to the models.
AI analysis is not available	Continue image review and human decisions when appropriate. Ask the operator to check Model API readiness and the workstation connection.
No PRISM overlays	Check Show AI suggestions , the lesion filter, and whether a result is available. An empty result does not prove lesion absence.
Draft save failed	Keep the page open, correct the reviewer or connectivity issue, and choose Save draft again.
Case is not DR-ready	Check image admission, final grade, reviewer, timestamp, and grade provenance.
Case is not Lesion-ready	Check the active annotation set and whether its hash changed after Confirm Annotation .
CVAT is unavailable	Continue local review where possible. Retry the optional integration after the configured URL, project, credentials, and network are available.

7.2 What the workstation records

The workstation records source and analysis identity, patient and eye context, image admission state, clinician decisions, reviewer names, timestamps, annotation revisions, confirmation hashes, inference metadata, model provenance, and dataset readiness. The source image remains immutable.

7.3 Safety and privacy

WARNING - clinical use

This is a research and clinical-support proof of concept. Model output is assistive evidence. Human review remains authoritative. The product does not establish autonomous diagnosis, referral, treatment, or disease truth.

CAUTION - data handling

Use only approved public, synthetic, or authorized clinical data. Do not put PHI, private DICOM headers, tokens, passwords, CVAT credentials, model weights, or runtime databases in screenshots, examples, exports, or source control.

7.4 Glossary

Term	Meaning
AI suggestion	A model output shown to support inspection.
DR-ready	The case satisfies the existing dataset rules for the DR-grade task.
Lesion-ready	The case satisfies the existing dataset rules for lesion annotations.
Model score	A value returned by a model. It is not automatically a calibrated probability.
Original AI evidence	The class, score, geometry, identity, model, and provenance returned by inference before human edits.
Reviewer preference	A browser-local default name for new or unattributed review forms. It is not authentication.